

Variable selection - Workflow

1. Purpose

Goal. Build a relatively simple linear model with a small, data-selected set of pairwise interactions.

2. Fixed modeling choices

Item	Specification
Elastic Net alpha	a_f = 0.3
Elastic Net l1_ratio	l1_f = 0.05
Estimator	scikit-learn ElasticNet, max_iter = 100000, tol = 1e-6, selection = 'cyclic'
Penalty rule	The intercept is not penalized.
Hierarchy	H1 = OLS on mains; H2 = Elastic Net on interactions to predict H1 residuals. Held-out prediction: y_hat = y_hat_H1 + r_hat_H2. If K = 0, omit H2 and set y_hat = y_hat_H1.
Redundancy cutoff	r_cut = 0.80

3. Definitions and conventions used in every stage

3.1 Data

- Outcome: Log2T (column name 'Log2T').
- Main predictors: defined by columns in the input file (example: sw5_input.csv – first column are the row labels, last column is the outcome Log2T).
- Sample size: defined by the input file (n = 76 independent observations).

3.2 Performance metric

Outer LOOCV pooled out-of-fold R². LOOCV yields one held-out prediction y_hat_i for each sample i. Compute

$$R2_OOF = 1 - [\text{sum}_i (y_i - y_hat_i)^2] / [\text{sum}_i (y_i - y_bar)^2]$$

where y_bar is the mean of observed y across all n samples. Do not compute fold-level R² values and average them.

3.3 Leak-free scaling and feature construction

- Within each current training set (outer-train, inner-train, or bootstrap resample), compute the mean mu_j and standard deviation sigma_j for each main predictor X_j using training rows only.
- Z-score mains as
 $Z_j = (X_j - \mu_j) / \sigma_j$.
- If sigma_j = 0 in a split, treat X_j as constant: set Z_j = 0 for all rows in that split, exclude that column from H1 in that split, and record the event.

- Construct all pairwise interactions from Z-scored mains as $W_{jk} = Z_j \times Z_k$ for all pairs $j < k$.
- Name each interaction deterministically as $j \times k$ (for example, F0xSL1).
- Within the same training set, compute μ_{jk} and σ_{jk} for each interaction W_{jk} , then Z-score interactions as $W_{\hat{jk}} = (W_{jk} - \mu_{jk}) / \sigma_{jk}$.
- If $\sigma_{jk} = 0$ in a split, drop that interaction from that split, set it to 0 for held-out rows, and record the event.
- Never compute any scaling parameter from held-out data.

Implementation note. The leak-free rule applies everywhere: outer folds, inner folds, and bootstrap resamples.

4. Workflow

Stage 1A— one screening table computed once on the full dataset

Goal: obtain overview statistics and performance for mains and all interactions. Get Interaction Stability Score for each interaction.

- Run LOOCV across all n samples; there is no outer loop at this stage.
- In each split, apply leak-free scaling on the inner-train set and construct all mains and all interactions.
- Fit H1 (OLS on Z-scored mains) on the inner-train set and compute residuals r on the inner-train set.
- For each interaction m , compute $ISS_m = |\text{corr}(r, W_{\hat{m}})|$ on the inner-train set only.
If $\text{var}(r) = 0$ or $\text{var}(W_{\hat{m}}) = 0$, set $ISS_m = 0$ for that split and record it.
- Fit H2 (Elastic Net on all interactions) on the inner-train set to predict r using fixed $(a_f, l1_f)$. Record, for each interaction, the nonzero indicator ($\beta_m \neq 0$) and coefficient sign when $\beta_m \neq 0$.
- Aggregate results across LOOCV splits to obtain the global mean $\pm$ SD ISS, nonzero frequency, and sign stability for every interaction.
- Add to the same diagnostic table the maximum absolute Pearson correlation to any other predictor (main or interaction), along with the identity of that predictor.

Stage 2A — Select a small interaction set from global Stage 1A diagnostics

Goal: Choose a stable, small interaction set using the single global Stage 1A table. K is selected by inner LOOCV performance with fixed $(a_f, l1_f)$.

- Define one global stable pool S_{global} from the GLOBAL Stage 1A table. Keep interaction m if
 - $\text{nonzero_freq}_m \geq 0.60$ and
 - $\text{sign_stability}_m \geq 0.80$
 Rank S_{global} by decreasing global mean ISS and let $M = |S_{\text{global}}|$.
- Apply the redundancy filter to S_{global} : remove any interaction whose absolute Pearson correlation exceeds r_{cut} with either a main variable or a higher-ranked interaction already retained in S_{global} .
- Choose K by inner LOOCV on the full dataset with fixed hyperparameters. For each K in $\{0, \dots, M\}$, define $*$ as the top- K interactions from S_{global} , run LOOCV with H1 followed by H2 using l_K , and compute pooled $R2_{\text{OOF}}(K)$.
Implementation note: reuse each LOOCV fold's standardized matrices instead of rebuilding them every time

- Apply the parsimony rule: choose the smallest K such that $R2_OOF(K)$ is within 0.01 of the best observed $R2_OOF$. This K is K^* with the interaction set I_K^* .
- Set $I_fixed = I_K^*$. Evaluate I_fixed by outer LOOCV: in each outer fold, fit H1 and H2 on the outer-train set using I_fixed and fixed ($a_f, I1_f$), predict the held-out sample, and compute pooled outer $R2_OOF$.

Stage 2B — conditional bootstrap coefficient stability ($B = 100$)

- Condition on I_fixed and the fixed hyperparameters.
- For each bootstrap replicate, resample rows, compute scaling on the bootstrap sample only, fit H1 and H2, and record standardized coefficients.
- Summarize coefficients by mean, SD, and 2.5th and 97.5th percentiles across bootstrap replicates.

Stage 3A — direct Elastic Net on mains + I_fixed

- Refit a single Elastic Net model on all mains and I_fixed with fixed ($a_f, I1_f$); do not separate H1 and H2 in this step.
- Compute LOOCV pooled $R2_OOF$.
- Run conditional bootstrap ($B = 100$) on the same main-effect and interaction set, using bootstrap-sample-only scaling, and summarize standardized coefficients by mean, SD, and 2.5th and 97.5th percentiles.

5. Reporting checklist

Report all outputs below as one .excel file. Combine tables where practical, but do not omit content. Include intercepts wherever applicable.

- Summary table with pooled outer LOOCV $R2$ for all stages, number of interactions, hyperparameters used, and any filters applied.
- K-versus- $R2$ curve.
- Stage 1A diagnostics table: ISS, nonzero frequency, and sign stability reported as mean $\pm$ SD across folds.
- One table for Stage 2A and 2B: include Stage 2A selected K^* , I_fixed , the $R2_OOF(K)$ curve, report outer LOOCV $R2_OOF$ for the fixed-set model
- For Stages 2A and 2B, report standardized coefficients as mean $\pm$ SD across folds. In the same table include bootstrap summaries ($B = 100$): coefficient mean $\pm$ SD and percentile intervals.
- For Stages 3A, report standardized coefficient mean $\pm$ SD across folds. In the same table include bootstrap summaries ($B = 100$): coefficient mean $\pm$ SD and percentile intervals. Include for each main and interaction a column listing alternative variables, i.e. mains or interactions with Pearson correlation $|R| \geq 0.80$.